INCH-POUND

MIL-DTL-32811 (MR)
4 July 2026

DETAIL SPECIFICATION

TITANIUM: SHAPES, FORGED OR FORMED; BARS, FLATS, AND SPECIAL SECTIONS (FOR ARMOR)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers weldable titanium alloy bars, flats and special sections produced from billets to specified dimensions by forging or hot forming, for armored vehicle components.

1.2 Classification. Forged or Formed Titanium Armor (FTA) is aligned with MIL-DTL-46077 titanium armor plate classes and is of the following classes as specified (see 6.2 and table II).

1.2.1 Class I. Titanium shapes, forged or hot formed corresponding to MIL-DTL-46077, class 1 which is an Extra Low Interstitial (ELI) grade (see table I). If no specific class is ordered on the contract or purchase order, the default class is class I.

1.2.2 Class II. Titanium shapes, forged or hot formed corresponding to MIL-DTL-46077, class 2 which is an alpha-beta titanium alloy, specifically, 6Al-4V with intermediate oxygen levels ranging from greater than 0.14 to 0.20 weight percent (see table I and 6.1).

1.2.3 Class III. Titanium shapes, forged or hot formed corresponding to MIL-DTL-46077, class 3 which permits oxygen levels beyond 0.20 weight percent suitable for applications such as appliqué where maximum ductility and toughness are not necessary (see table I).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.2.4 Class IV. Titanium shapes, forged or hot formed corresponding to MIL-DTL-46077, class 4 alloy compositions consisting of Al, V, & Fe chemistries outside the conventional Ti-6Al-4V chemistries.

1.2.5 Class IVa. Titanium shapes, forged or hot formed corresponding to MIL-DTL-46077, class 4a alloy compositions. Class IVa specific chemistry requirements are listed in table I.

1.2.6 Class IVb. Titanium shapes, forged or hot formed corresponding to MIL-DTL-46077, class 4b alloy compositions. Class IVb specific chemistry requirements are listed in table I.

1.3 Geometries. The titanium should be of the following shapes and sizes as specified (see 6.2):

a. Bars, round: 1/4 inch up to and including 4 inch diameter.
b. Bars, square: 1/4 inch up to and including 4 inch.
c. Flats, 1/4 inch or more in thickness: Not over 6 inches wide.
d. Shapes of special sections: In accordance with applicable drawings.

2. APPLICABLE PUBLICATIONS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-46077	-	Armor Plate, Titanium Alloy, Weldable

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product
MIL-STD-3081	-	Arc Welding of Armor Grade Titanium

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/).

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ASTM INTERNATIONAL

ASTM B348/B348M	-	Standard Specification for Titanium and Titanium Alloy Bars and Billets
ASTM E8/E8M	-	Standard Specification for Tension Testing of Metallic Materials
ASTM E539	-	Standard Test Method for X-Ray Emission Spectrometric Analysis of 6Al-4V Titanium Alloy
ASTM E1409	-	Standard Test Method for Determination of Oxygen and Nitrogen in Titanium and Titanium Alloys by the Inert Gas Fusion Technique
ASTM E1447	-	Standard Test Method for Determination of Hydrogen in Titanium and Titanium Alloys by Inert Gas Fusion Thermal Conductivity/Infrared Detection Method
ASTM E2371	-	Standard Test Method for Analysis of Titanium and Titanium Alloys by Direct Current Plasma and Inductively Coupled Plasma Atomic Emission Spectrometry (Performance-Based Test Methodology)
ASTM E2994	-	Standard Test Method for Analysis of Titanium and Titanium Alloys by Spark Atomic Emission Spectrometry and Glow Discharge Atomic Emission Spectrometry (Performance-Based Method)

(Copies of these documents are available online at https://www.astm.org).

SAE INTERNATIONAL

AMS 2249	-	Chemical Check Analysis Limits - Titanium and Titanium Alloys
AMS 2380	-	Approval and Control of Premium-Quality Titanium Alloys
AMS 4903	-	Titanium Alloy Sheet, Strip, and Plate, 6Al - 4V, Solution Heat Treated
AMS 4904	-	Titanium Alloy Sheet, Strip, and Plate 6Al - 4V, Solution Heat Treated and Aged
AMS 4911	-	Titanium Alloy, Sheet, Strip, and Plate, 6Al - 4V,Annealed
ARP 982	-	Minimizing Stress Corrosion Cracking in Wrought Titanium Alloy Products

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General requirements. Titanium shapes shall be in accordance with the applicable drawings, the requirements of this specification, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.4).

3.2 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required, and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for ballistic acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and ballistic acceptance test samples or sections shall be manufactured by the process proposed for use on production items. The prepared articles for ballistic acceptance shall be submitted to Commander, USA ATC, ATTN: TEDT-AT-SL- V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059. No first article requirements shall be waived without review and approval by the procuring contracting officer (see 3.16 and 6.4).

3.2.1 First time producer. First time producers wishing to qualify to this specification shall follow the instructions of 6.6.

3.3 Chemical composition. A declared chemistry shall be submitted to the contracting agency or its authorized representative. The chemical composition of the declared chemistry shall conform to the requirements of table I. The first article and the production test samples shall utilize the same declared chemistry within the allowable ranges proposed for use in production. A statement showing the heat analysis of each melt and one product analysis of each lot, complete details of the heat treatment, and all mechanical properties of each lot of production articles shall be furnished for the files of the purchaser at no cost. All elements of the chemical composition specified in table I shall be shown in the statement. Any additives or hardening agents intentionally added shall be declared. The supplier may establish and submit separate limits for each thickness of armor to be furnished.

3.3.1 Class compositions. The chemical composition of the articles shall be as specified in table I for classes I, II, III, IVa, and IVb. If no class is specified on the contract, drawing or order, class I shall be supplied. Chemical composition variances shall meet the requirements of SAE AMS 2249.

3.3.1.1 Class IV.

3.3.1.1.1 Class IV composition. Upon approval of the procuring activity, an alternate composition of titanium alloy may be authorized. In addition to the composition limits specified in table I, suppliers must submit to the procuring activity established limits for each intentionally added element as well as maximum limits for others.

3.3.1.1.2 Class IV density. The density (see 6.9) shall be 0.158 through 0.165 pounds per cubic inch (4.373 g/cm^3 through 4.567 g/cm^3). Density may be calculated based on chemical composition range established.

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3.3.1.2 Class IVa. This specific composition of a class IV titanium alloy has been tested and approved as an alternative composition of a class IV titanium alloy.

3.3.1.3 Class IVb. This specific composition of a class IV titanium alloy has been tested and approved as an alternative composition of a class IV titanium alloy.

TABLE I. Chemical composition - percent by weight. 1/

Class	Al	V	C	O	N	H 2/	Fe	Ti 3/	Other 4/	
									Each	Total
I	5.50 - 6.50	3.50 - 4.50	0.04	0.14	0.02	0.0125	0.25	Rem	0.10	0.40
II	5.50 - 6.75	3.50 - 4.50	0.08	0.20	0.05	0.0150	0.30	Rem	0.10	0.40
III	5.50 - 6.75	3.50 - 4.50	0.08	0.27	0.05	0.0150	0.30	Rem	0.10	0.40
IV	5/	5/	0.08	0.27	0.05	0.0150	5/	Rem	5/	5/
IVa	3.5 - 4.5	2.0 - 3.0	0.08	0.27	0.03	0.0150	1.2 - 1.8	Rem	0.10	0.30
IVb	4.2 - 5.4	2.5 - 3.5	0.08	0.15 - 0.19	0.03	0.0150	0.5 - 0.7	Rem	0.10	0.30

1/ Where single units are shown, these indicate the maximum amounts permitted.
2/ Hydrogen shall be determined on each lot of the product as shipped.
3/ Titanium is determined by differences.
4/ Other elements need not be analyzed nor reported unless otherwise specified.
5/ For class IV alternate composition details see 1.2.4 and 3.3.1.1.1.

3.4 Heat treatment. The supplier shall select the heat treatment of the material to meet the requirements of this specification. A statement of the heat treatment to be used shall be submitted to the procuring contracting officer. All bars, flats or special sections in each lot, including samples, shall receive the same heat treatment except for such variations in tempering temperature as may be necessary to produce the prescribed mechanicals. The finishing temperature shall be reported on the Formed Titanium Armor (FTA) Test Data Form (see figure 1).

3.5 Reheating. Unless otherwise approved by the procuring activity (see 6.2), after the final heat treatment, material shall not be reheated or processed above the final annealing, stress relieving, or aging temperatures or to a temperature that is 100° F below the beta transus, whichever is less. If any articles are retreated above these limiting temperatures, they shall be so recorded and completely re-tested as a separate lot.

3.5.1 Repairing. Weld repairs after the final heat treatment, whether completed before or after shipment from the production facility, must be specifically authorized by the procuring contracting officer (see 6.2) and performed in accordance with MIL-STD-3081 by the procuring contracting officer.

3.5.2 Hot forming. “Unless otherwise authorized by the procuring activity, when hot forming is performed in conjunction with or after the final heat treatment, the material shall not be reheated or processed above the final annealing, stress relieving or aging temperatures or to a temperature that is 100F below the beta transus, whichever is less. If plates are hot formed above these limiting temperatures, they shall be so recorded and completely re-tested as a separate lot”

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3.6 Quality. The quality and cleanliness of bars, flats, or special sections up to and including 64 square inches in cross sectional area shall be equal to or better than the relevant levels provided in ASTM B348/B348M under the Workmanship, Finish, and Appearance section. When greater levels of quality beyond ASTM B348/B348M are required, the procuring activity may refer to AMS 2380 for additional examination criteria and procedures.

3.7 Production weldability. To ensure production weldability, the supplier must adhere to the following, unless otherwise directed by the contract, purchase order, or drawings:

a. Demonstrate Capability: The titanium bars, flats, or special sections must be weldable using standard production methods.
b. Gain Approval: This is proven by submitting sample production welds to the procuring contracting officer for approval, prepared in accordance with MIL-STD-3081.
c. Waiver: The procuring contracting officer has the discretion to waive the sample requirement if they approve of the material's known welding characteristics.

3.8 Mechanical properties. The minimum mechanical properties requirements for each of the classes appear within table II of MIL-DTL-46077 listing the corresponding classes.

3.8.3 Tensile requirements for special sections. For special sections, the average yield strength and elongation of all specimens shall be within 10% of the average yield strength and 5% of the average elongation unless otherwise specified on the drawing or in the contract or purchase order. The yield strength and elongation readings shall be reported on the FTA Test Data Form (see figure 1).

3.8.4 Test blocks for special sections. When the special section dimensions do not permit direct sampling from which to produce mechanical specimens, a test block of sufficient volume shall be prepared following the thermomechanical process of the special section it represents to accommodate the preparation of mechanical test specimens upon approval of the procuring activity.

3.9 Ballistic requirements. Ballistic requirements per critical thickness or thicknesses as established within the contract or purchase order shall be in accordance with Appendix A of MIL-DTL-46077 specification as applicable. A ballistic test article must be of the same melt of titanium, of the same thickness, having the same treatment, and heat treated using the same thermal cycle process in the same production heating and cooling equipment in the same facility. When a formed part is ordered, one possible method of meeting this requirement is to use a cutout or drop section of a completed formed part. When specified by the procuring activity (see 6.2), production materials and ballistic test articles shall be allowed to be heat treated separately. To facilitate the correct usage of the MIL-DTL-46077, Appendix A ballistic acceptance requirements, cross referencing between the MIL-DTL-32811 classes the corresponding wrought titanium plate classes are provided in table II with additional guidance for the default class noted. Procuring activities may establish alternate ballistic requirements as required.

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TABLE II. MIL-DTL-32811 relationship to MIL-DTL-46077 wrought titanium plate.

MIL-DTL-32811 Formed Armor Class	MIL-DTL-46077 Wrought Plate Class
I [1/]	1
II	2
III	3
IV	4
IVa	4a
IVb	4b

[1/] Class I is the default when no class for MIL-DTL-32811 is indicated in the engineering drawing, contract or purchase order.

3.9.1 Partial penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect: when four (4) partial penetrations are above the minimum requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be specified) above the minimum requirement.

3.9.2 Ballistic test articles greater than 4 inches ordered thickness. Ballistic testing of forged or formed components over 4 inches ordered thickness is only required if requested by the procuring activity. Additional allowances for mounting (see 3.9.2.1 and 4.3.6) may be needed and are authorized to accommodate complex shapes, larger components, or geometries.

3.9.2.1 Integral temporary tabs for handling of larger components. When specified in the contract, purchase order, or engineering drawing, larger components requiring heavy machinery to move or manipulate may use temporary integral removable tabs, lobes, or lugs to facilitate handling, lifting, positioning, machining, and ballistic test mounting as part of the design of the forged or formed component. When used, these temporary tabs shall be designed for placement in non-critical areas where their presence in the forged or formed component and subsequent removal by cutting and grinding operations shall not degrade the structural integrity, final functional geometry, and the ballistic response of completed component. Preparation of lifting holes (see 3.9.2) for larger components greater than 4 inches for ballistic test articles can be waived when temporary tabs are present and in usable positions.

3.9.3 Ballistic testing of forged or formed components. The frequency of ballistically testing production forged or formed components shall be specified in the applicable engineering drawing contract or purchase order (see 6.2).

3.9.4 Ballistic test article information. For each lot, a properly completed FTA Test Data Form (see figure 1) shall be submitted with each ballistic test article or section that represents that particular processing lot.

3.10 Dimensions and dimensional tolerances.

3.10.1 Dimensions. Bars, flats, and special sections shall comply with the dimensions as specified on the applicable drawings or specified in the contract or purchase order.

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3.10.2 Tolerances. The thickness of any bars or flats, after final heat treatment, shall not vary by more than the amounts indicted in ASTM B348/B348M. Tolerances for special sections shall be as specified in the contract or purchase order.

3.10.2.1 Tolerances for special sections. Tolerance for special sections shall be according to the engineering drawing, contract, or purchase order.

3.11 Identification. Identification markings and records shall be such as to ensure positive identification of all bars, flats and special sections, samples and specimens, with the lot and corresponding heat from which they were produced. The key to identification symbols shall be furnished to the inspector prior to submittal of material for inspection and testing.

3.12 Physical marking for identification. Each bar, flat, and special section or its section or surrogate shall be legibly marked in at least one location in characters not less than 0.5 inch high with the following information.

(a) Manufacturer's name, trademark, or both.
(b) Number of this specification.
(c) Class.
(d) Army part or drawing number.
(e) Lot number.

3.12.1 Ballistic test samples. First article, acceptance, and ballistic test articles shall be legibly marked with the information in 3.12. In addition, first article bars, flats, and special sections shall be marked "PRE" and acceptance test bars, flats, and special sections marked "ACC". This marking shall be marked in at least two opposing locations in letters not less than 0.5 inch high unless otherwise directed by the engineering drawing, contract, or purchase order.

3.13 Information required. In addition to the information required in the FTA test data form (see figure 1), a statement showing the heat analysis of each heat and complete details of the heat treatment of each lot shall be furnished for the files of the procuring activity. All elements of the chemical composition shall be shown in the statement, including special additives or hardening agents, whether shown in table I or not.

3.14 Information required on ballistic test samples. Unless otherwise specified in the contract or purchase order, a properly completed FTA Test Data Form (see figure 1) shall be submitted with each ballistic test shape or section containing the information as specified in MIL-STD-662. Ballistic retest articles shall be marked "R1" and "R2" respectively (see Appendix A rejection and retest procedures from the corresponding wrought plate specifications).

3.15 Workmanship. Bars, flats, and special sections shall be free from injurious imperfections including cracks, pits, porosity, laminations, segregation, and inclusions, for their intended use.

3.16 Waiver of first article test. Suppliers who have previously produced forged or formed bars, flats, and special sections for armor and have not significantly changed production practices and meeting all the requirements of this specification in the past 37 months, will not be required to perform first article tests.

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REQUEST FOR BALLISTIC TEST OF FORMED TITANIUM ARMOR

SPECIFICATION: MIL-DTL-32811	**REVISION:**	**AMENDMENT:**
FIRING RECORD:	**DATE:**	**TEST CONTRACT NO:**

	SOURCE PLATE	**FORGE**	**HEAT TREATER**
Name:			
Address:			
POC:			
Phone No:			
Email:			

Test Item Identification: _____	Thickness (max): _____ Thickness (min): _____	Heat Treatment: _____ Finishing Temperature: _____
Lot No._____ Forging No._____	FTA Classification CLASS: _____ Carbon Equivalence (CE): _____	Forging Repaired (y/n):_____ Remarks:_____

PURPOSE:___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing

SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)

CHEMISTRY ANALYSIS:

Al	V	Fe	O	C	N	H	Si			Ti
										Rem

MECHANICAL PROPERTIES:

1: Longitudinal Axis	UTS (ksi): _____	0.2% YS (ksi): _____	Elongation (%):_____
2: Transverse Axis	UTS (ksi): _____	0.2% YS (ksi): _____	Elongation (%):_____

TENSILE TESTS FOR SPECIAL SECTIONS ONLY

YIELD STRENGTH (ksi) 1: _____ **2:** _____ **3:** _____ **4:** _____ (special sections only)

ELONGATION (%) 1: _____ **2:** _____ **3:** _____ **4:** _____ (special sections only)

BALLISTIC TEST RESULTS: SEPARATELY HEAT TREATED TEST ARTICLE: YES ____ **NO** ____

	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (ft/sec)**	**Actual V_{50} (ft/sec)**	**Pass/ Fail**
Test:					
Projectile:					

MIL-DTL-46077 V_{50} CLASS USED: _____

DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS

Lot [met] [failed to meet] the ballistic requirements of specification MIL-DTL-32811

Government Representative:	Date:	Supplier Representative:	Date:

FIGURE 1. Formed Titanium Armor (FTA) Test Data Form.

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4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.1.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Lot. Unless otherwise indicated in the contract or purchase order, a lot shall consist of all production and ballistic test bars, flats, and special sections of the same shape, chemistry, having the same treatment, and heat treated using the same thermal cycle process in the same production heating and cooling equipment in the same facility submitted for inspection as a unit, provided that the lot does not exceed 300 forgings or formed parts. When specified by the procuring activity (see 4.2.1 and 6.2), production and ballistic test articles shall be allowed to be heat treated separately.

4.2.1 Separately heat treated ballistic test article. When the procuring activity allows a ballistic test article to be heat treated separately from the production articles it represents (see 4.3), it shall be so stated in the FTA Test Data Form (see figure 1).

4.3 First article inspection. First article inspection, except as contraindicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4. First article inspection shall consist of the following:

a. Visual examination (see 3.10, 3.12, 3.14, and 4.3.1).
b. Dimensional confirmation of near net shape (see 3.10).
c. Chemical analysis (see 3.3, table I, 4.3.3, 4.5.1.1, and 4.6.1).
d. Titanium Quality (see 3.6 and 4.3.4).
e. Production weldability (see 3.7 and 4.3.5).

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f. Mechanical properties (see 3.8 and 4.3.6).
g. Ballistic testing (see 3.9, table II, 4.3.7, 4.7, and 6.6).

4.3.1 For visual examination. Unless a sampling plan is specified by the procuring contracting officer, all bars, flats or special sections comprising the lot shall be subject to visual inspection to determine compliance with the identification marking (see 3.12) and workmanship (see 3.15) requirements.

4.3.2 Initial dimensional verification. Bars, flats or special sections selected in accordance with 4.3.1 shall be dimensionally interrogated to ensure the initial dimensions and tolerances achieved are sufficient from which to final machine the ordered component (see 3.10).

4.3.3 For chemical analysis. A sample for chemical analysis (see 4.5.1.1 and 4.6.1) shall be taken from each heat.

4.3.4 For titanium quality. Unless otherwise specified (see 6.2), samples for titanium quality tests shall be taken representing the top, middle and bottom of the first, middle and last usable ingots of each heat. Samples shall be sufficiently large to prepare test specimens required in 4.5.1.2.

4.3.5 For production weldability. When required, the number and size of samples shall be as specified or approved by the procuring contracting officer and welded in accordance with MIL-STD-3081.

4.3.6 For mechanical properties. Samples for mechanical properties tests of bars, flats, or special sections shall be taken at random with frequency in accordance with the contract or purchase order (recommendations are in 6.2.1). Special sections sampling shall be as required in 4.3.6.1. Mechanical property samples shall be of the same thickness or diameter as the bars, flats, or special sections they represent. The samples shall be either as large as the parts represented or shall be of sufficient size to allow for the number of specimens required in 4.5.1.3 to be taken from them. All specimens shall be taken so that every part of the test specimen will be outside of the heat affected zone of any thermally cut edge.

4.3.6.1 Samples for mechanical properties for special sections. Unless otherwise indicated in the contract or purchase order, samples for mechanical properties of special sections including tensile (see 3.8.3) shall be taken from the thickest and thinnest areas of the article.

4.3.7 For ballistic testing. Unless otherwise indicated in the contract, purchase order, or engineering drawing, for each configuration that the manufacturer has under contract, one or more of the bars, flats, or special sections shall be submitted for ballistic testing in accordance with 3.9 and Appendix A of the MIL-DTL-46077 wrought titanium armor plate specification (see 1.2 and table II). Prior to the submission of the bars, flats, or special sections, the procuring activity, the tester, and the contractor will reference and discuss a drawing of the articles and the ballistic test conditions: namely, the component thickness, impact area, and projectile type. The number of articles needed for first article ballistic acceptance testing will then be determined by the procuring activity. MIL-DTL-46077 and its corresponding class used for ballistic acceptance testing shall be reported in the space provided on the FTA Test Data Form (see figure 1). Modifications, cutting, welding, and bracketing will be determined as needed for test mounting and for lifting and manipulation of thicker components (see 3.9.2).

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4.4 Quality conformance (acceptance) inspection. Test for acceptance of a single lot shall consist of testing all requirements specified in section 3. For a series of lots, subsequent verification conformance inspection shall consist of the following:

a. Visual examination (3.12, 3.14, and 4.3.1).
b. Dimensional (see 3.10).
c. Ballistic testing (see 3.9, 4.3.7, and 4.7).
d. Preparation for shipment (see 5.1).

4.4.1 Visual. Bars, flats, or special sections selected in accordance with 4.3.1 shall be visually examined for compliance with identification marking (see 3.12, 3.14) and workmanship (see 3.15).

4.4.2 Dimensional. Bars, flats or special sections selected in accordance with 4.3.1 shall be dimensionally examined for compliance with dimensions and tolerances (see 3.10).

4.4.3 Ballistic testing. From each lot, one bar, flat or special section shall be submitted for ballistic testing (see paragraph 4.3.7 for supplementary information).

4.4.4 Preparation for shipment. Examination shall be made to determine compliance with the requirements for packaging (see 5.1).

4.5 Tests.

4.5.1 Test specimens.

4.5.1.1 For chemical analysis. A sample sufficient to perform the analysis (see 4.6.1) required to determine compliance with 3.3 and table I shall be selected from each lot.

4.5.1.2 For titanium quality tests. Test specimens to verify titanium quality shall be prepared as specified in AMS 2380.

4.5.1.3 For mechanical properties. At least two tension test specimens shall be taken from each lot to corresponding to the thickest and thinnest sections of the formed article. The tension test specimens shall conform to the largest obtainable round cross-section specimen shown in ASTM E8/E8M. The location and type of specimens shall be as specified in 4.3.6 and 4.3.6.1.

4.5.1.3.1 Tensile specimens for special sections. For special sections, at least four tensile specimens shall be taken from each sample. Two specimens shall each be removed from each special section in the thickest and thinnest regions. Actual location shall be as specified on the drawing or in the contract or order. All specimens shall be in the same orientation with respect to maximum metal flow. Specimen geometries shall be the largest dimensions possible while in compliance with ASTM E8/E8M. If the special section size is too small for multiple tensile specimens, additional articles shall be furnished to meet the total specimen quantity.

4.6 Test methods.

4.6.1 Chemical analysis. One sample shall be selected from each heat, and first article chemical composition requirements shall be determined using ASTM E539 X-ray emission spectrometric

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method for aluminum, vanadium and iron. Oxygen shall be analyzed by the inert gas fusion technique specified in ASTM E1409. Hydrogen shall be analyzed by the inert gas fusion thermal conductivity method specified in ASTM E1447. Carbon and nitrogen shall be determined using ASTM E2371. Other analytical methods such as ASTM E2994 may be used with the written approval of the acquiring activity. In case of a dispute, the results of chemical analysis by the ASTM E2371 method shall be used as the basis for acceptance or rejection. The analysis shall comply with the declared composition established in accordance with the requirements of table I (see 3.3).

4.6.2 Titanium quality tests. Titanium quality tests shall be in accordance with ASTM E348.

4.6.3 Production weldability. When weldability tests are required, the test methods and acceptance criteria shall be specified or approved by the procuring contracting officer and shall be in accordance with MIL-STD-3081.

4.6.4 Mechanical properties tests. Conformance to the mechanical property requirements shall be determined by testing at least two specimens (one longitudinal and one transverse) in the condition ordered from each sample selected in accordance with 4.5.1.3. Tension tests shall be accomplished in accordance with the following provisions.

4.6.4.1 Tension specimen configuration. The tension specimens shall conform to the following configuration in accordance with ASTM E8/E8M:

a. For product thickness up to 0.375 inch, standard "sheet-type" rectangular cross-section tension specimens shall be used.

b. For product thickness equal to or greater than 0.375 inch, standard round cross-section tension specimens with a 0.250 inch diameter minimum reduced test section shall be used.

4.6.4.2 Tension specimen preparation. Unless the lot consists of only one unit, not more than one specimen per direction per test condition shall be taken from each sample. Longitudinal designated specimens shall be prepared in such a manner that the longitudinal axis of the specimen is parallel to the principal major axis of the forged or formed sample. Transverse designated specimens shall be prepared in such a manner that the longitudinal axis of the specimen is perpendicular to the major axis direction and the thickness direction of the sample.

4.6.4.3 Tensile tests for special sections. First article tensile tests for special sections shall be conducted in accordance with ASTM E8/E8M.

4.6.4.4 Tension test method. Tension tests shall be accomplished in accordance with ASTM E8/E8M, except that the strain rate shall be limited to 0.003 to 0.007 inch/inch per minute for the measurement of the yield strength: after that it is to be increased so as to produce tensile failure in approximately one additional minute. In case of dispute, the results of referee tension tests performed on a tensile machine having a strain pacer and using a strain rate of 0.005 inch/inch per minute through the yield strength and a minimum crosshead speed of 0.10 inch per minute when the yield strength has been exceeded shall govern.

4.7 Ballistic testing of production forged or formed titanium shapes. When specified on the drawing or contract (see 6.2), ballistic testing of production forged or formed titanium shapes to the critical thickness or thicknesses region(s) on the shape as established within the contract or

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purchase order (see 3.9 and 4.3.7) shall be conducted by the government at a government test facility specified in the contract on order. Ballistic testing shall be conducted in accordance with the requirements of Appendix A of the MIL-DTL-46077 wrought plate specification corresponding to the MIL-DTL-32811 formed titanium class cited in table II unless otherwise indicated in the contract or purchase order. Procuring activities may establish alternate ballistic requirements as required.

4.8 Rejection and retests.

4.8.1 Rejection. Where one or more test specimens fail to meet the requirements of this specification the lot represented by the specimen or specimens shall be subject to rejection.

4.8.2 Resubmitting new first article test sample. To obtain first article approval, the supplier shall submit a new first article test sample for each sample which fails to meet the requirements of this specification.

4.8.3.1 Non-ballistic retests of bars, flats or special sections. All failed sample units from the original sample shall be excluded from the retest lot. Four new sample units shall be taken and tested in accordance with 4.6.1, 4.6.2, 4.6.3, and 4.6.4 as applicable. Failure of any retest specimen to meet the applicable property requirements shall be cause for rejection of the entire lot and no further retesting shall be permitted.

4.8.4 Chemical analysis retest and rejection. Chemistry retest and rejection shall be in accordance with ASTM E2371.

4.8.5 Ballistic retests of production bars, flats and special sections. When a production bar, flat or special section fails to meet the ballistic test requirements of the corresponding wrought plate Appendix A, the bar, flat or special section involved shall be subject to rejection. Failure of any bar, flat or special section to meet the ballistic test requirements of the relevant Appendix A to the ordered class for this specification shall require the selection of the next bar, flat or special section manufactured following notification of failure under the ballistic test.

4.8.5.1 Shipping special sections from the producing facility. No special sections of the die(s) involved shall be shipped from the producing facility until the first otherwise acceptable special section has passed the ballistic tests (see 3.9).

4.8.5.2 Ballistic retesting of a lot represented by the bars, flats or special sections exhibiting ballistic performance below the requirements of the corresponding Appendix A. At the option of the contractor, a lot represented by the bars, flats or special sections exhibiting ballistic performance below the requirements of the aligning Appendix A, may be retested ballistically after reheat treatment.

4.9 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced, and providing the manufacturer agrees in writing to maintain the same manufacturing procedures. The manufacturer when eligible for reduced or audit testing may at their option, continue to submit one ballistic test sample or section per lot of material. The test

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facility shall only test the non-required test samples or sections upon manufacturer's loss of reduced testing eligibility.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles. Class I is equivalent to class 1 material specified by previous revisions of MIL-A-46077, which is an Extra Low Interstitial (ELI) grade. Class II is equivalent to class 2 MIL-DTL-46077 material specified by AB-1 composition originally defined under the cancelled MIL-T-9046J and SAE AMS-T-9046 specifications and still referenced within numerous engineering drawings and contracts. These legacy specifications were replaced by three SAE AMS specifications based upon the following heat treated conditions:

Annealed	AMS 4911
Solution Treated	AMS 4903
Solution treated and aged (STA)	AMS 4904

Class III based upon MIL-DTL-46077 class 3, permits a higher oxygen level which would be suitable for applications where maximum ductility and toughness are not necessary, such as appliqué. Class IV based upon class 4 of MIL-DTL-46077 permits compositions different from the conventional Ti-6Al-4V alloy and should be used as appliqué armor only. A weldability study should be conducted prior to any welding of Class 4 armor. Classes IVa and IVb correspond to classes 4a (ATI 425) and 4b (Timet 108) respectively from MIL-DTL-46077.

The titanium classes covered by this specification are military unique. The bars, flats and special sections are intended for use as components on armored combat vehicles, emplacements and the like for protection against armor piercing projectiles, bursting shell and fragments of high explosive ammunition. When authorized by the procuring activity, further fabrication or repair by welding in accordance with MIL-STD-3081 may be involved.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Specify titanium shape, size, and class (see 1.2 and 1.3).
c. When first article sample is required (see 3.2).
d. If the declared chemistry composition can be different (see 3.3).
e. Heat treatment to be used if required (see 3.4).
f. Where samples for titanium quality tests should be taken if other than in 4.3.4.
g. If ballistic testing of production components over 4" is required (see 3.9.2).
h. If ballistic forged or formed shapes are authorized to be heat treated separately (see 3.9, 4.2, and 4.2.1).
i. If weld repair after final heat treatment is authorized (see 3.5.2).
j. If alternate sampling is indicated (see 6.2.1.1).
k. Packaging requirements (see 5.1).

6.2.1 Frequency of sampling. In the absence of a sampling plan defined in the contract or purchase order, table III includes the default frequency of sampling.

TABLE III. Default frequency of sampling.

Length as heat-treated	No. of samples
Greater than 12 feet	1 from each 25 items or 20,000 lbs 1/
Greater than 2 to 12 feet incl. or 100 lbs. or less	1 from each 50 items or 10,000 lbs 1/
2 feet or less or 10 lbs. or less	1 from each 100 items or 5,000 lbs 1/

1/ Whichever is preferred.

6.2.1.1 Alternate sampling. When alternate sampling is indicated, or requested, the procuring activity should follow MIL-STD-1916 and ANSI/ASQ Z1.4 guidance.

6.3 First article. When first article inspection is required, the procuring contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first (how many) production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The procuring contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Waivers or deviations. The procuring contracting officer should coordinate all requests for waiver or deviation to this specification (see 3.16 and 4.9) with the appropriate design and product assurance activities.

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6.5 Chemical analysis. Suggested ASTM instrumental methods that can be used for chemical analysis are E539, E1409, E1447, E2371 and E2994.

6.6 Potential suppliers. Potential suppliers who have not previously supplied forged or formed armor shapes to this specification and wish to have their material ballistically tested, may do so at their own expense. It is recommended that inquiries for such testing be directed to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance. The armor production process should be inspected by a cognizant government representative to ensure compliance with the details of this specification.

6.7 Alignment of MIL-DTL-32811 classes to MIL-DTL-46077 armor plate specification. To ensure the correct requirements are used for any ordered class of MIL-DTL-32811 forged or formed armor shapes, the designers, producers, suppliers, and procurement officials should carefully verify with 1.2 and table II to ensure the full and correct requirements including Appendix A ballistic acceptance data from MIL-DTL-46077, "Armor Plate, Titanium Alloy, Weldable" for ballistic acceptance are followed in preparation for the contract or purchase order. The wrought titanium armor plate specification contains multiple classes and additional attention to detail should be made to ensure the correct class is selected and documented in the FTA Test Data Form (see figure 1) or as otherwise directed in the contract or purchase order.

6.8 Stress corrosion. Certain processing procedures and service conditions may cause these titanium alloy products to be subject to stress corrosion cracking. SAE ARP 982 recommends practices to minimize the susceptibility to such stress corrosion cracking.

6.9 Relationship of mechanical properties to ballistic requirements. The minimum mechanical properties specified may not assure titanium armor meeting the specified ballistic requirements.

6.10 Density. The density of the specified classes listed in this specification are as follows:

Class I (Ti-6Al-4V) = 0.16 lb/in^3 (4.43 g/cm^3).

Class II (Ti-6Al-4V) = 0.16 lb/in^3 (4.43 g/cm^3).

Class III (Ti-6Al-4V) = 0.16 lb/in^3 (4.43 g/cm^3).

Class IV (range) = 0.158 lb/in^3 through 0.165 lb/in^3 (4.373 g/cm^3 through 4.567 g/cm^3).

Class IVa (ATI 425) = 0.161 lb/in^3 (4.456 g/cm^3).

Class IVb (Timet 108) = 0.161 lb/in^3 (4.456 g/cm^3).

6.11 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors listed in table IV.

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TABLE IV. Conversion factors. [1/]

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

[1/] Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide.”

6.12 Subject term (key word) listing.

Ballistic	Combat vehicles	Minimum Acceptance
Billets	Extra Low Interstitial (ELI) grade	Vehicles

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
Army - MR
(Project FORG-2026-002)

Review activities:
Army - AR, AT, AV, TE

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